ESD Tcl commands

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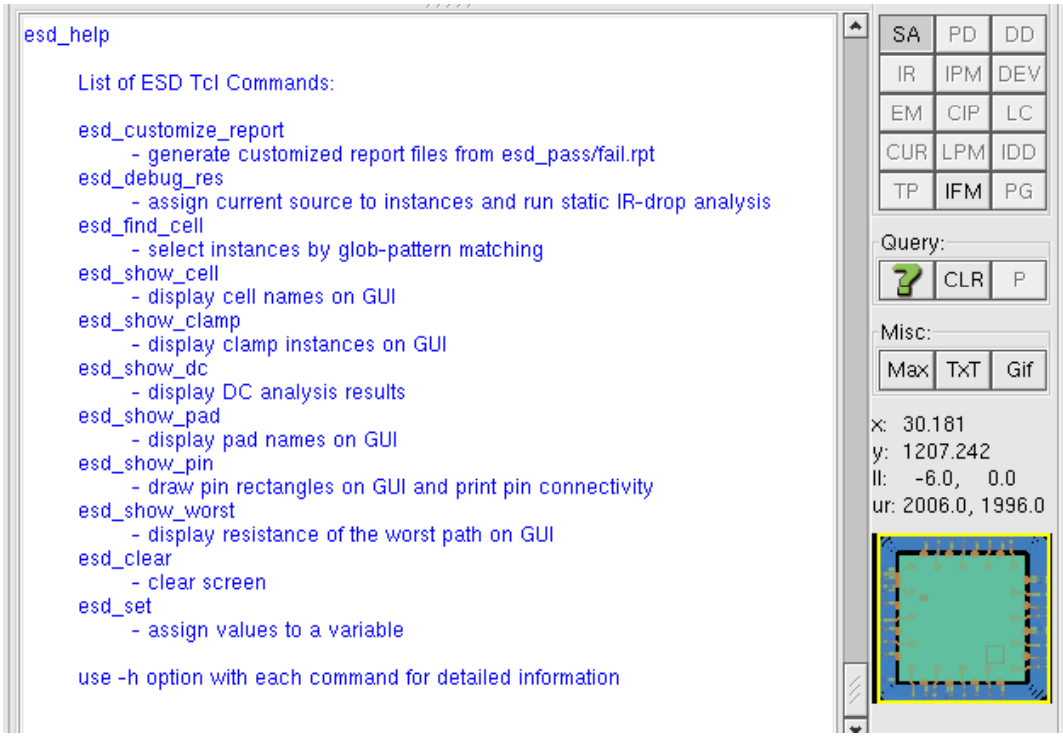
Loading esd.tbc

1. source $env(APACHEROOT)/scripts/esd.tbc [enter]
   - ✓ Valid in RH10.1.1 or newer version

2. esd_help [enter]
   - ✓ To see available commands



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ESD Tcl commands

- **esd_customize_report**
  - customize esd pass/fail report files
- **esd_debug_res**
  - assign current source to clamp instances and run static IR drop analysis
- **esd_find_cell**
  - find cells or instances by glob-pattern matching
- **esd_select_inst**
  - sample instances at random for CLAMP2INST analysis
- **esd_show_cell**
  - display cell names or instance names
- **esd_show_clamp**
  - display PIN pair and Ron of clamp instances
- **esd_show_dc**
  - display I/V/R of clamp instances
- **esd_show_pad**
  - display pad names or find pads
- **esd_show_pin**
  - draw LEF pin rectangles and print pin connectivity
- **esd_show_worst**
  - display results one by one in order from worst to best
- **esd_clear**
  - clear screen
- **esd_set**
  - assign values to a variable

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esd_customize_report

- Purpose
  - ✓ To rearrange B2C report files on the basis of clamp instances, for INST2BUMP(core2bump) report generation
  - ✓ To generate INST2BUMP2CLAMP report files from two B2C results. (only accept pre-defined rule names, I2B2C_I2B and I2B2C_B2C)

- Usage

```
esd_customize_report
                -format/-f <I2B/I2B2C>
                ?-rule/-r <rule_names>?

                <FORMAT>
                I2B : generate INST2BUMP report files from BUMP2CLAMP report files
                I2B2C : generate INST2BUMP2CLAMP report files
                        from BUMP2CLAMP report files

        <usage>
                esd_customize_report -f i2b
                esd_customize_report -f i2b2c
```

- Note
  - ✓ 'esd_show_worst' can read I2B2C_pass/fail.rpt

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esd_debug_res (1/2)

- Purpose
  - ✓ To assign current to instances and run static IR analysis on the fly

- Usage
  - ✓ Click an instance and run 'esd_debug_res'
  - ✓ Current is distributed to all VIAs in the instances
  - ✓ With '-clamp' option, current is assigned to VIAs in the clamp pins

```
esd_debug_res
                ?-inst/-i <inst_names>?
                ?-current/-c <current>?
                ?-clamp? ?-rule/-r <rule_names>?

                -clamp : assign current to clamp pins
                -current : current value (default: 1A)

        <usage>
                esd_debug_res
                esd_debug_res -c 1m
                esd_debug_res -c 1m -clamp
```

- Note
  - ✓ To exclude specific pads, 'esd_show_pad –remove' can be used

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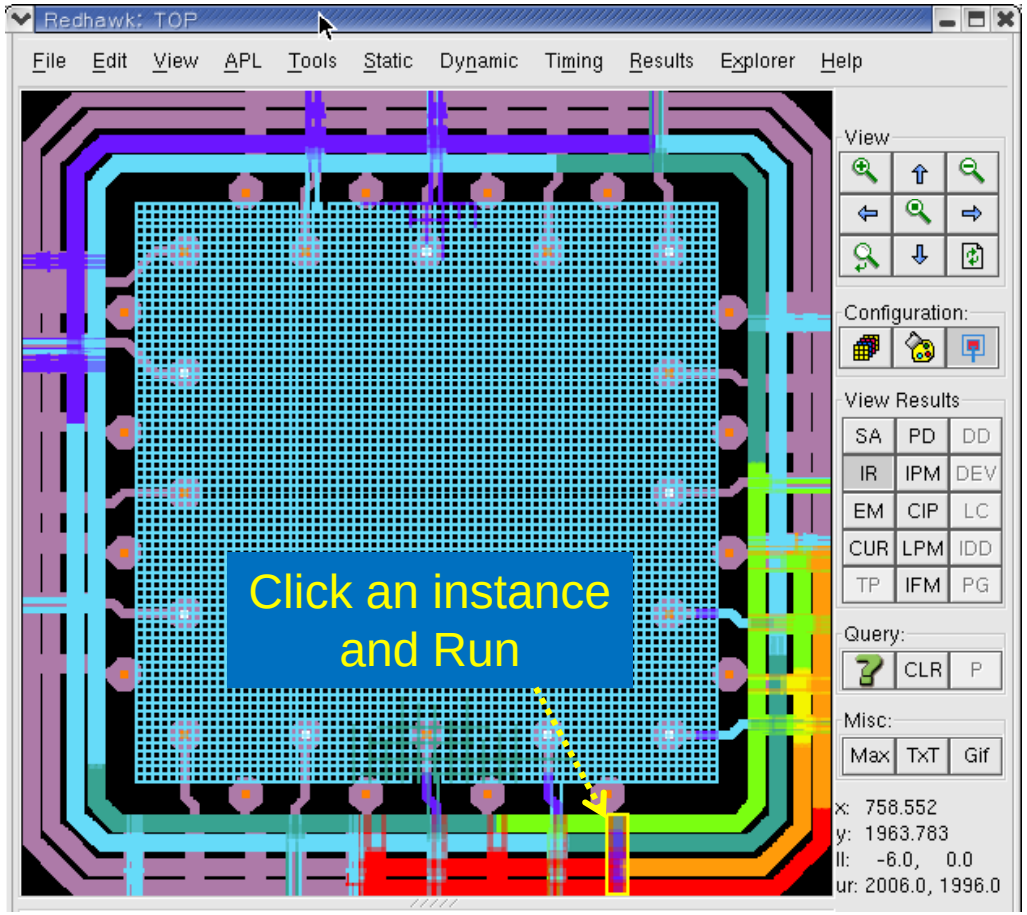
esd_debug_res (2/2)



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esd_find_cell (1/2)

- Purpose
  - ✓ Find cells or instances by glob-pattern matching

- Usage

```
esd_find_cell
                -cell/-c <cell_names> or -inst/-i <inst_names> or
                -cell_file/-cf <cell_list_files> or -inst_file/-if <inst_list_files>
                ?-legend/-l?
                ?-overlay/-ov?
                ?-iv?
                ?-o <output_file>?
                ?-all? or ?-limit <number>?

                -legend : show legend
                -overlay : preserve previous drawing
                -iv : make layers invisible for instance view
                -o : write selected instances to a file
                -all : display all instances (default limit 1000)

        <usage>
                esd_find_cell -c *FILLER* *BRK*
                esd_find_cell -i *DD* -l
```


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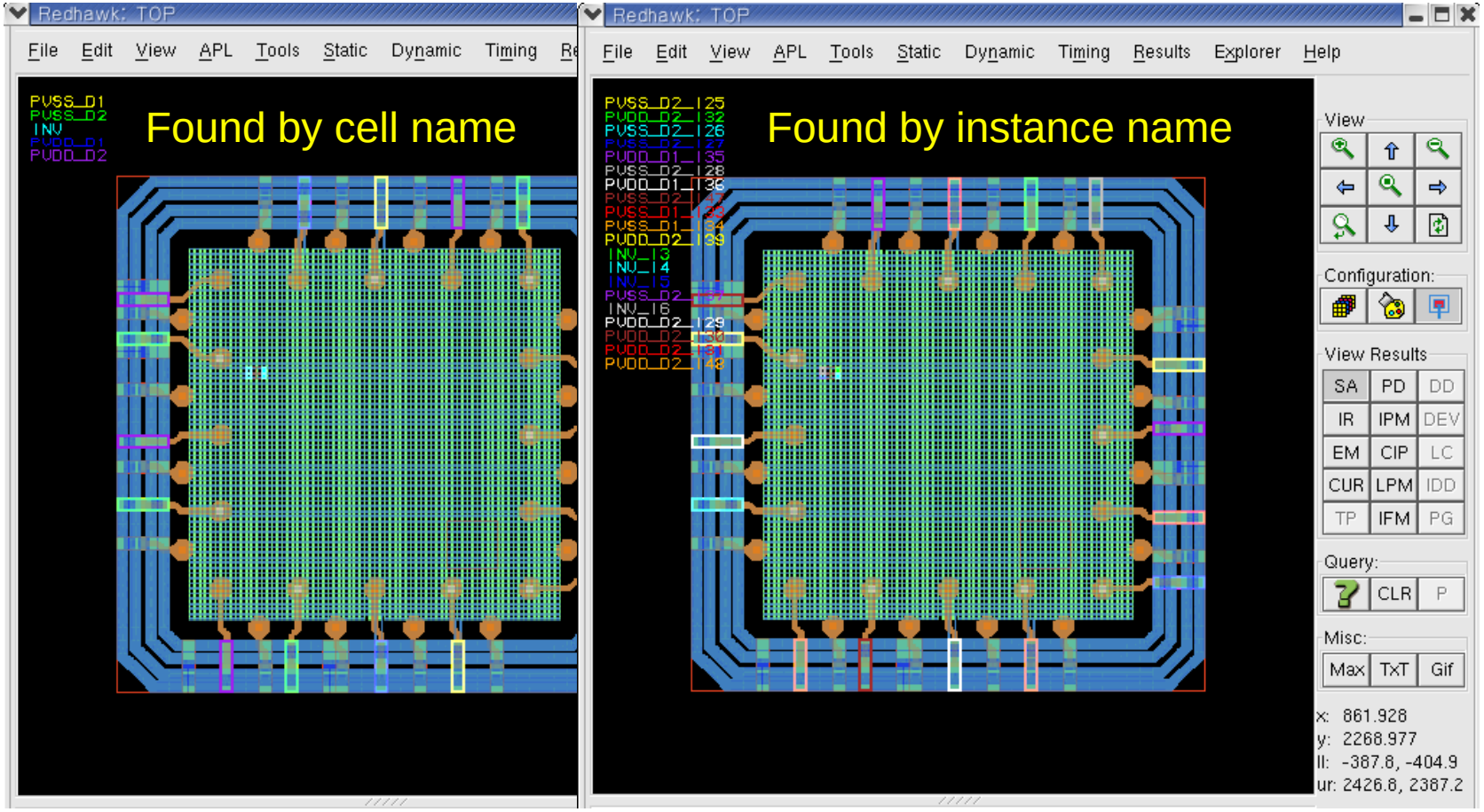
esd_find_cell (2/2)



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esd_select_inst (1/2)

- Purpose
  - ✓ Almost the same as atclSampleInst.
  - ✓ To show the progress of selection process, and provide more selection options

- Usage
  - ✓ Without '-n' option, it just shows partition. Use '-n' option to start
  - ✓ Use 'esd_select_inst -h' for more information

```
esd_select_inst
                -partition/-p <nX> <nY>
                -partition_size/-ps <sizeX> <sizeY>
                ?-n <nInst> or -n <value>%?
                ?-region <llx lly urx ury>?
                     :

                -partition : set partition number
                -partition_size : set partition size (in um)
                -n : number or percentage of instances per partition
                -limit : set limit of partition number (default 100)

        <usage>
                esd_select_inst -p 10 10 -n 10
                esd_select_inst -ps 500 500 -n 1%
```


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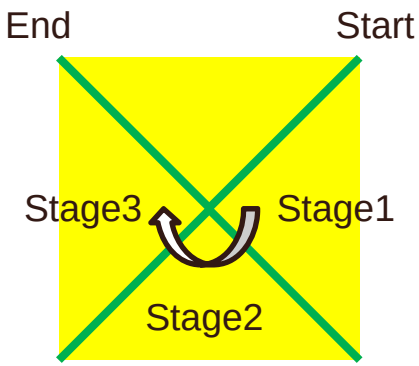
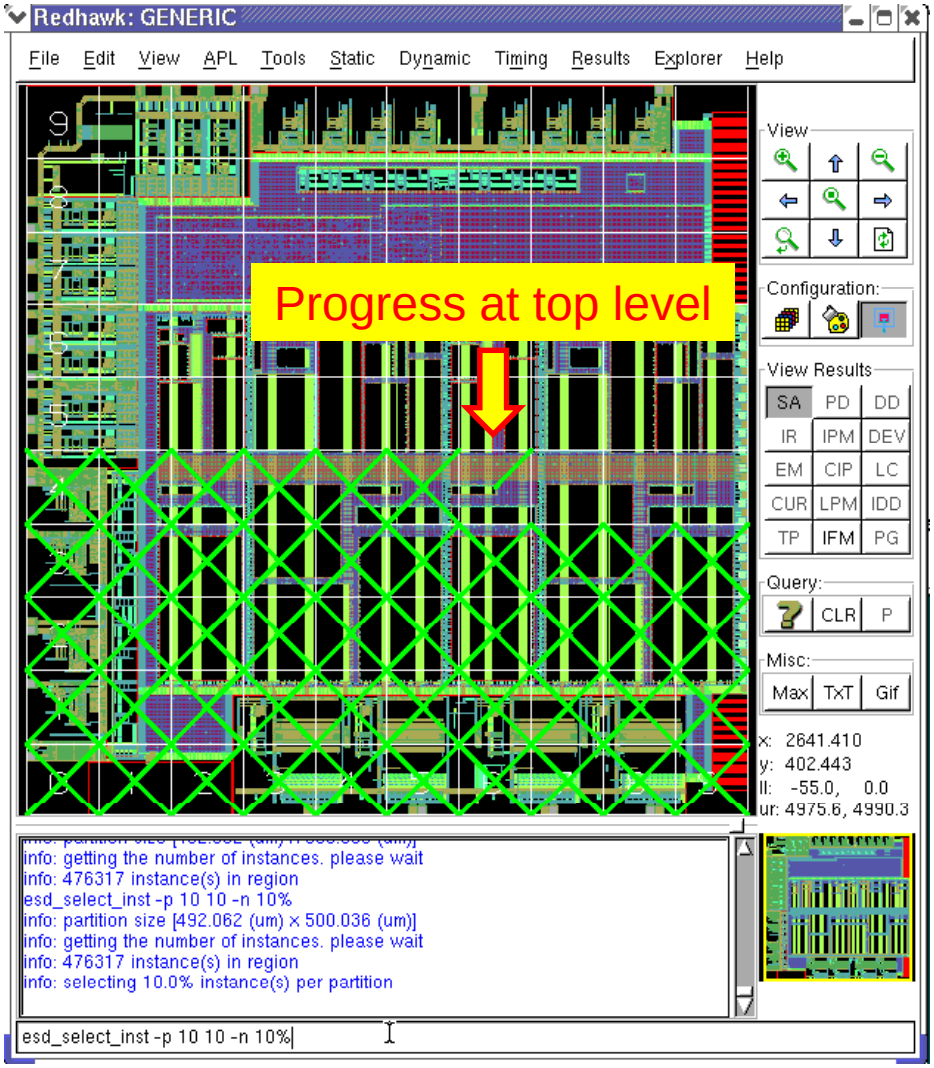
esd_select_inst (2/2)



Progress at partition level



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esd_show_cell (1/2)

- Purpose
  - ✓ As layout viewer, to show cell/instance placement

- Usage
  - ✓ '-size' option can be used to ignore standard cells

```
esd_show_cell
                ?-instance/-i?
                ?-overlay/-ov?
                ?-size <value in um>?
                ?-iv?
                ?-all? or ?-limit <number>?

                -instance : show instance names (default cell names)
                -overlay : preserve previous drawing
                -size : only consider larger cells than size x size
                -iv : make layers invisible for instance view
                -all : display all instances (default limit 500)

        <usage>
                esd_show_cell
                esd_show_cell -i
                esd_show_cell -size 10
```


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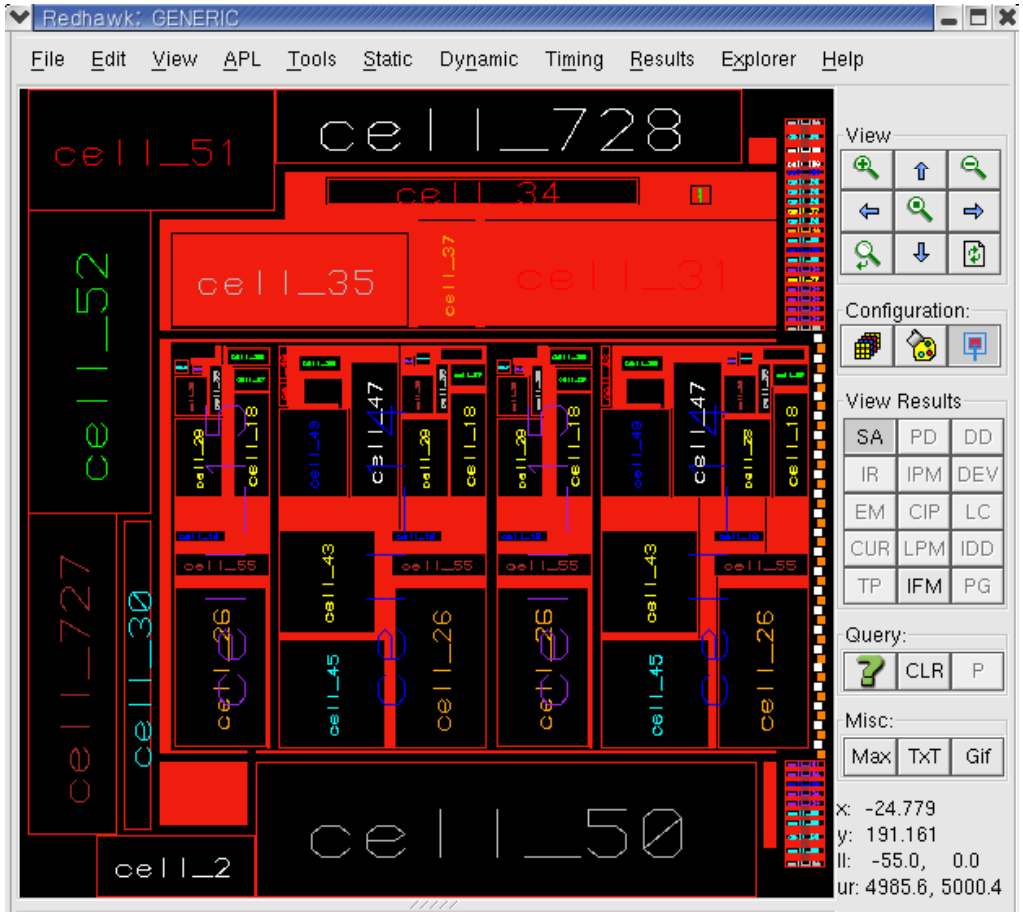
esd_show_cell (2/2)



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esd_show_clamp (1/2)

- Purpose
  - ✓ To show ESD_PIN_PAIR and Ron on each clamp instance by reading ClampInfo.rpt

- Usage

```
esd_show_clamp
                ?-rule/-r <rule_names>?
                ?-net/-n?
                ?-pin/-p?
                ?-overlay/-ov?
                   :

                -net : show net names (default locID and layer)
                -pin : show pin names (default locID and layer)
                -overlay : preserve previous drawing
                -iv : make layers invisible for instance view
                -all : display all clamp instances (default limit 500)

        <usage>
                esd_show_clamp
                esd_show_clamp -rule c2c c2m
```


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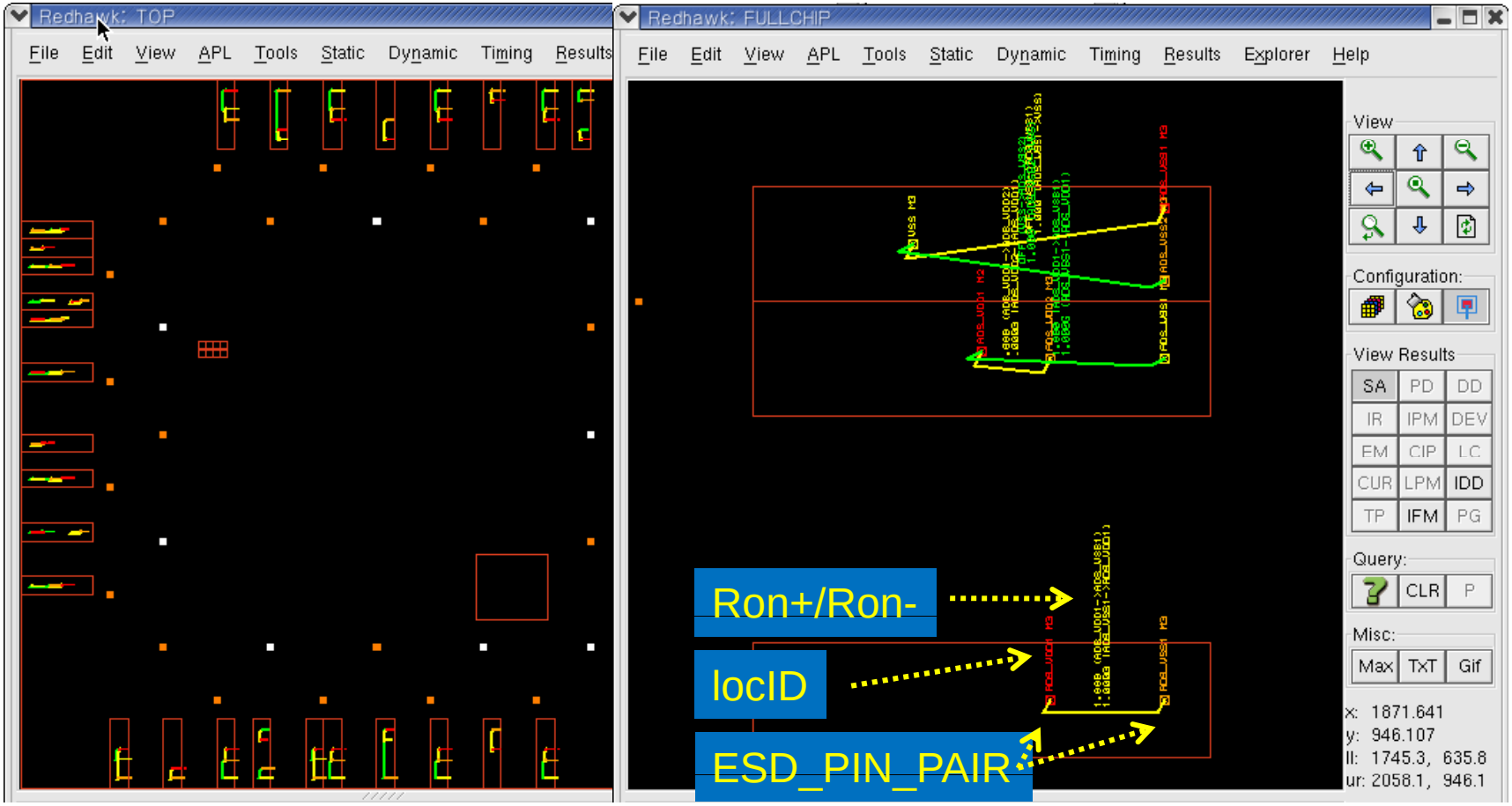
esd_show_clamp (2/2)



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esd_show_dc (1/2)

- Purpose
  - ✓ To shows current direction as well as I/V/R of each clamp instance by reading esd_summary.rpt

- Usage
  - ✓ Use '-h' option for more information

```
esd_show_dc
                ?-vol/-em/-cur?
                ?-imin/imax? ?-vmin/-vmax? ?-rmin/-rmax?
                ?-rule/-r <rule_names>?
                   :

                -vol/-em/-cur : display ewv/em/cur map
                -\[i|v|r\]min/-\[i|v|r\]max : display only clamps
                    between min and max
                   :

        <usage>
                esd_show_dc
                esd_show_dc –vol
                esd_show_dc –imin 1m –imax 100m
```


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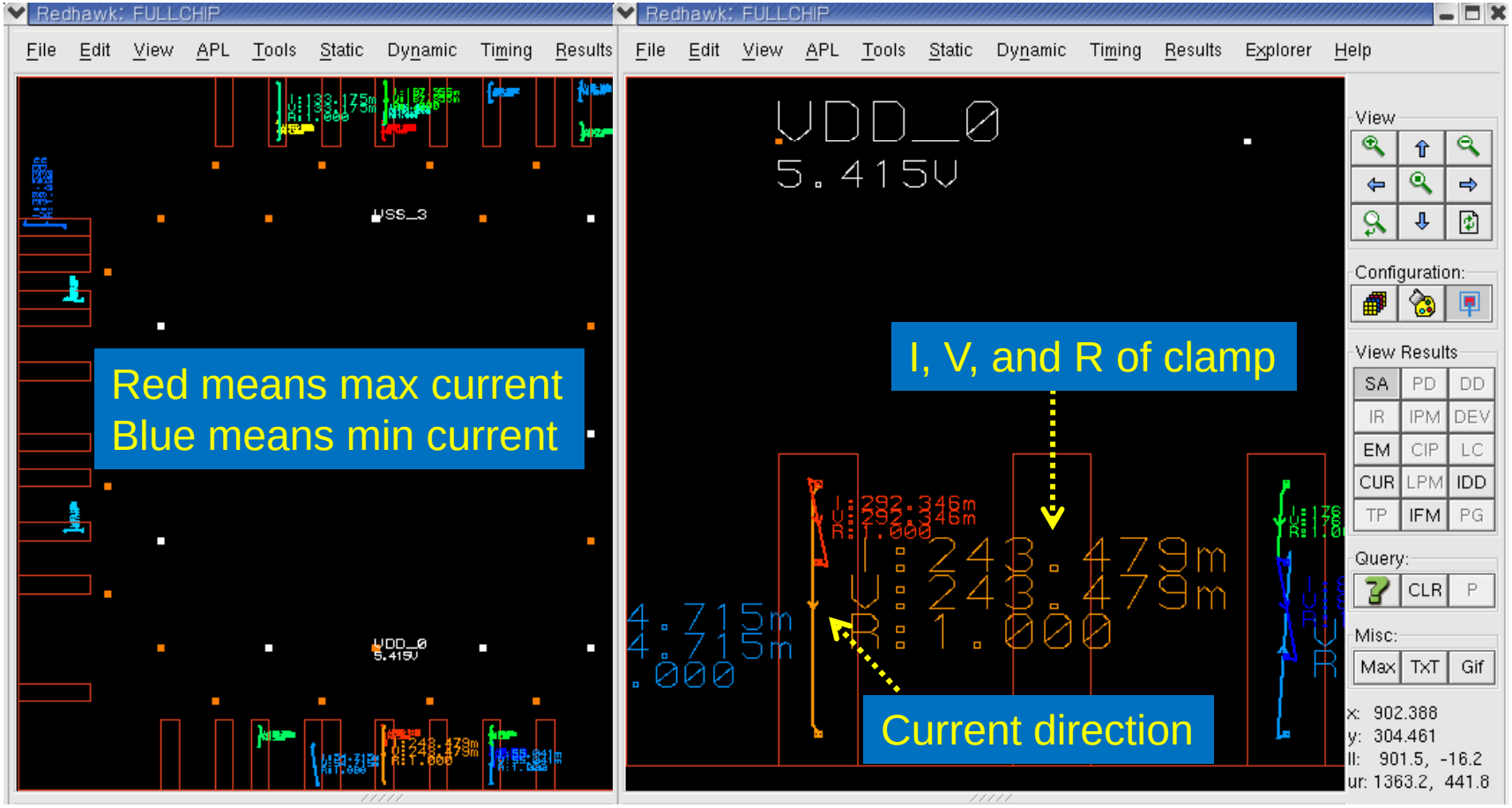
esd_show_dc (2/2)



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esd_show_pad (1/2)

- Purpose
  - ✓ To show pad names and find pads by glob-pattern matching

- Usage
  - ✓ '-ncc' option shows isolated bumps in B2B type analysis
  - ✓ '-remove' option can remove pads in current window

```
esd_show_pad
                ?-pad/-p <pad_names>?
                ?-net/-n <net_names>?
                ?-ncc? ?-rule/-r <rule_names>?
                ?-remove?
                    :

                -pad/-net : find pads by glob-pattern matching
                -ncc : show pads without clamps
                -remove : remove pads on screen
                    :

        <usage>
                esd_show_pad
                esd_show_pad -p VDD* *SS*
                esd_show_pad -n *DD* VSS*
                esd_show_pad -ncc -r b2bm
```


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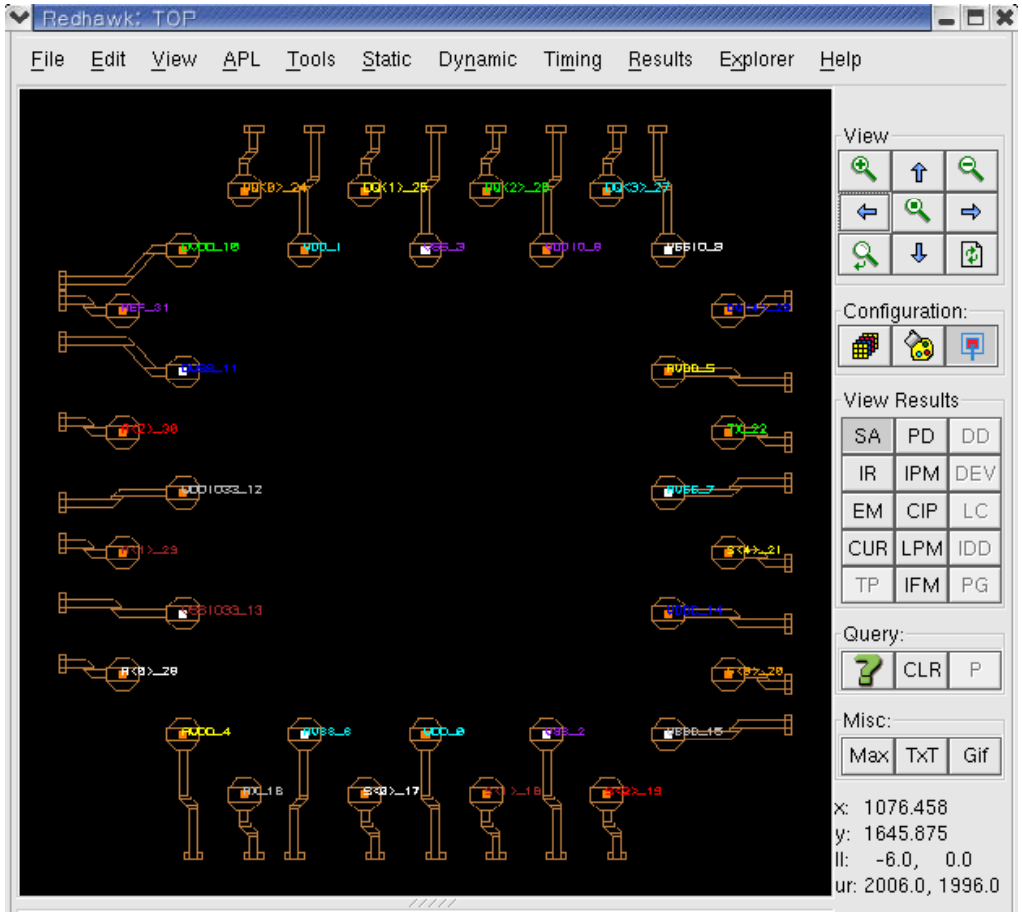
esd_show_pad (2/2)



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esd_show_pin (1/2)

- Purpose
  - ✓ To show LEF pin location on each instance, and check logical connectivity

- Usage
  - ✓ Click an instance and run 'esd_show_pin'
  - ✓ Use 'esd_show_pin -h' for more information

```
esd_show_pin
                ?-inst/-i <inst_names>?
                   :
                ?-conn_only/-co?
                ?-spt?

                   :
                -conn_only : print connectivity only
                -spt : perform min_res_path

        <usage>
                esd_show_pin
                esd_show_pin -co -spt
```


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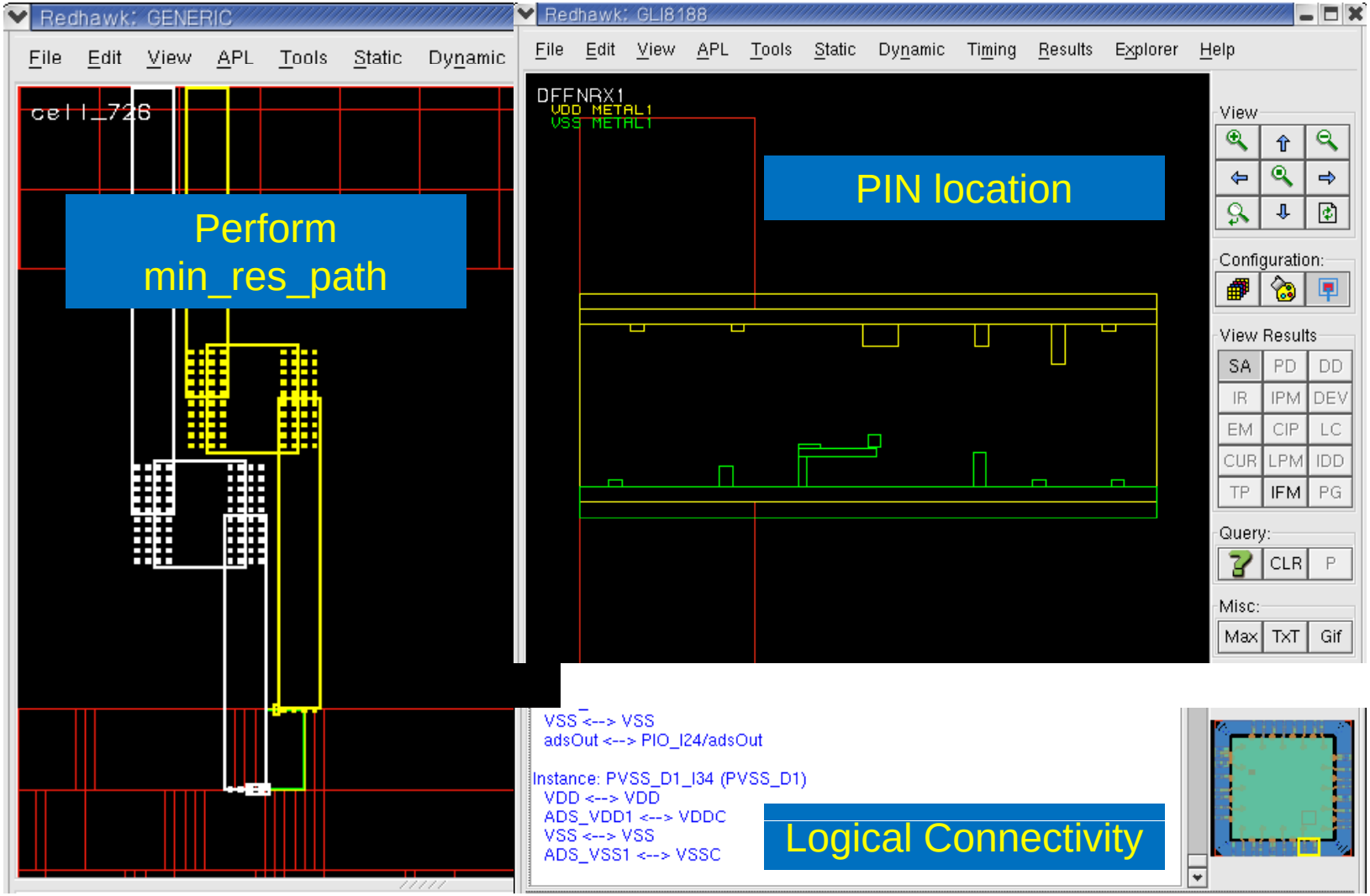
esd_show_pin (2/2)



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esd_show_worst (1/5)

- Purpose
  - ✓ To see analysis results one by one in order from worst to best

- Usage

```
esd_show_worst
                ?-rule/-r <rule_name>?
                ?-rank/-k <rank>/<from-to>?
                ?-net/-n <net_names>?
                ?-net_pair/-np <net1> <net2>?
                   :

                -rank : display n-th worst path
                -net : specify net names
                -net_pair : specify net pair for B2B/B2BM/C2I

        <usage>
                esd_show_worst
                esd_show_worst -n VDD VSS
                esd_show_worst -np VDD VSS
                esd_show_worst -r b2c -k 5
                esd_show_worst -r b2bm -k 1-10
```


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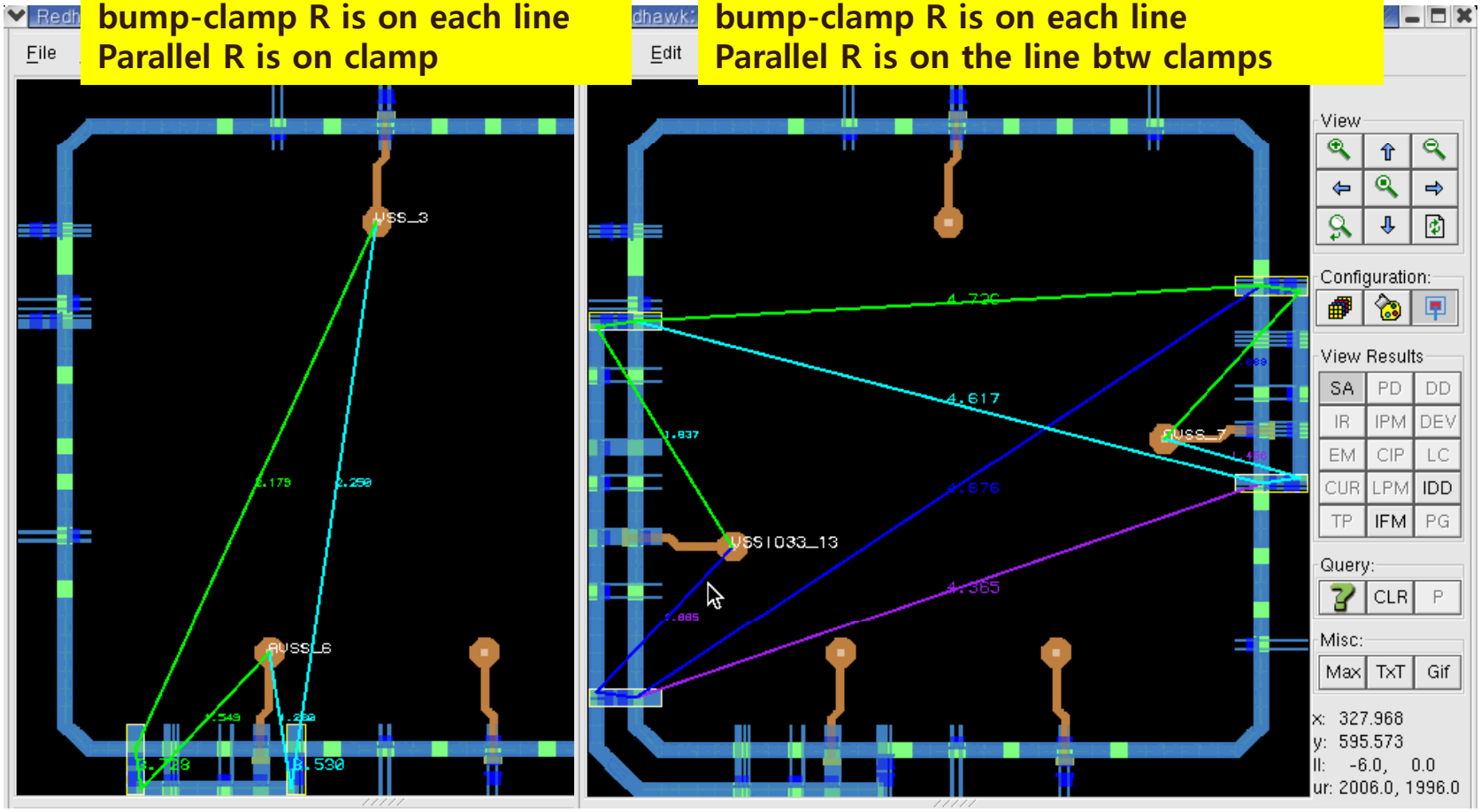
esd_show_worst (2/5)

B2B type
bump-clamp R is on each line
Parallel R is on clamp

B2BM type
bump-clamp R is on each line
Parallel R is on the line btw clamps



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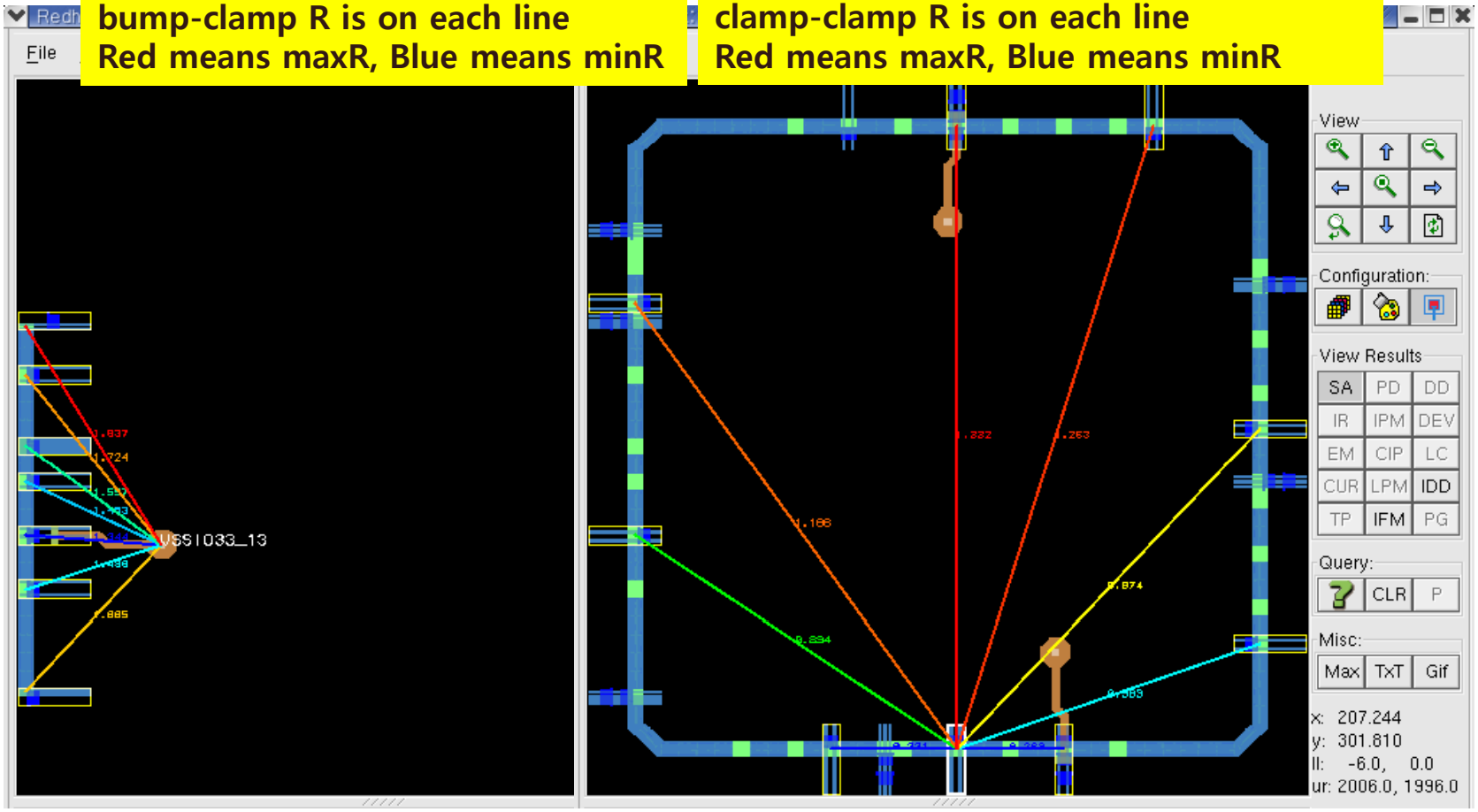
esd_show_worst (3/5)

B2C type
bump-clamp R is on each line
Red means maxR, Blue means minR

C2C Type
clamp-clamp R is on each line
Red means maxR, Blue means minR



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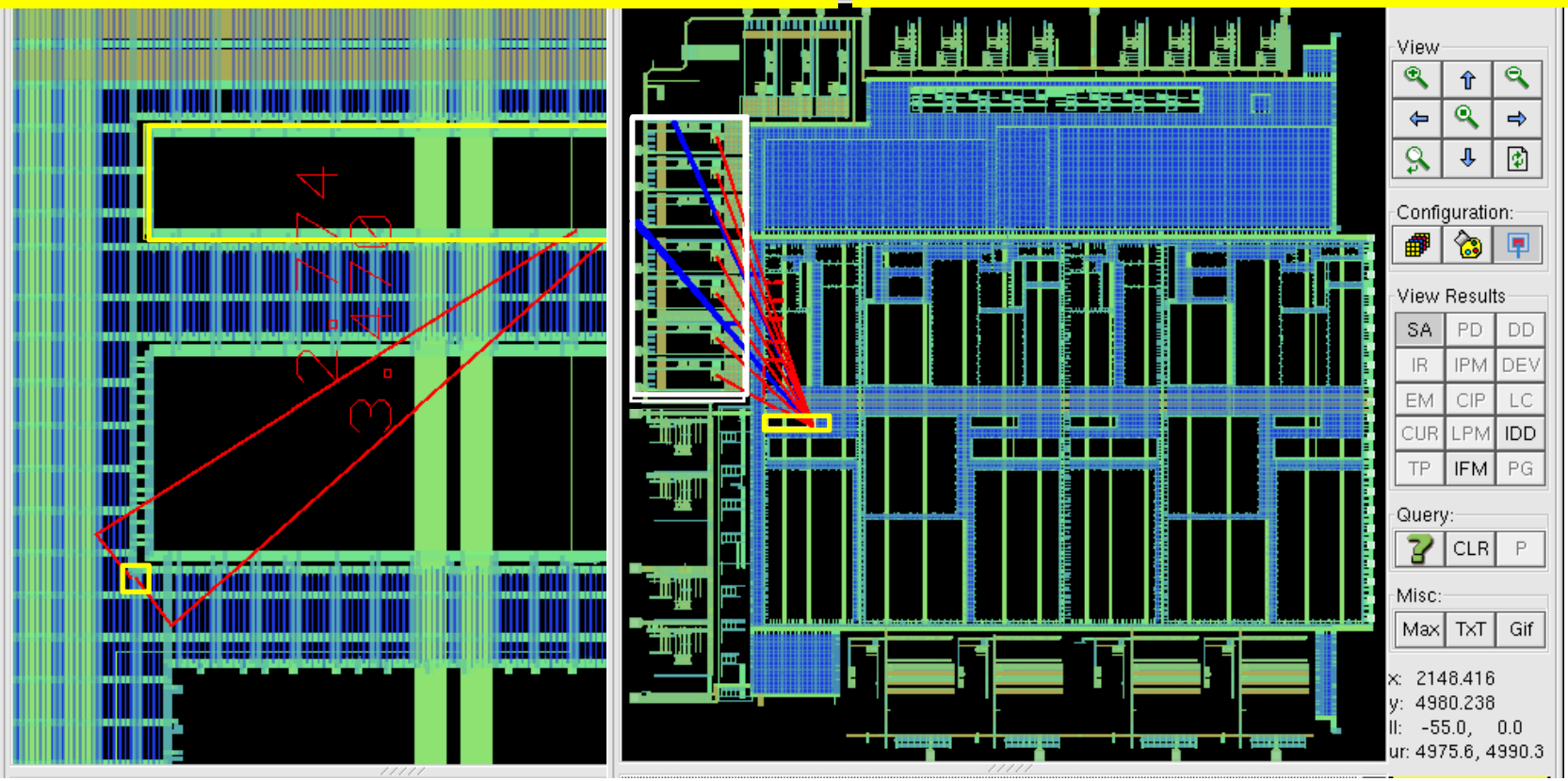
esd_show_worst (4/5)

C2I type
clamp-inst R is on each line
Red means max LoopR, Blue means min LoopR

C2M Type
clamp-macro R is on each line
Red means maxR, Blue means minR



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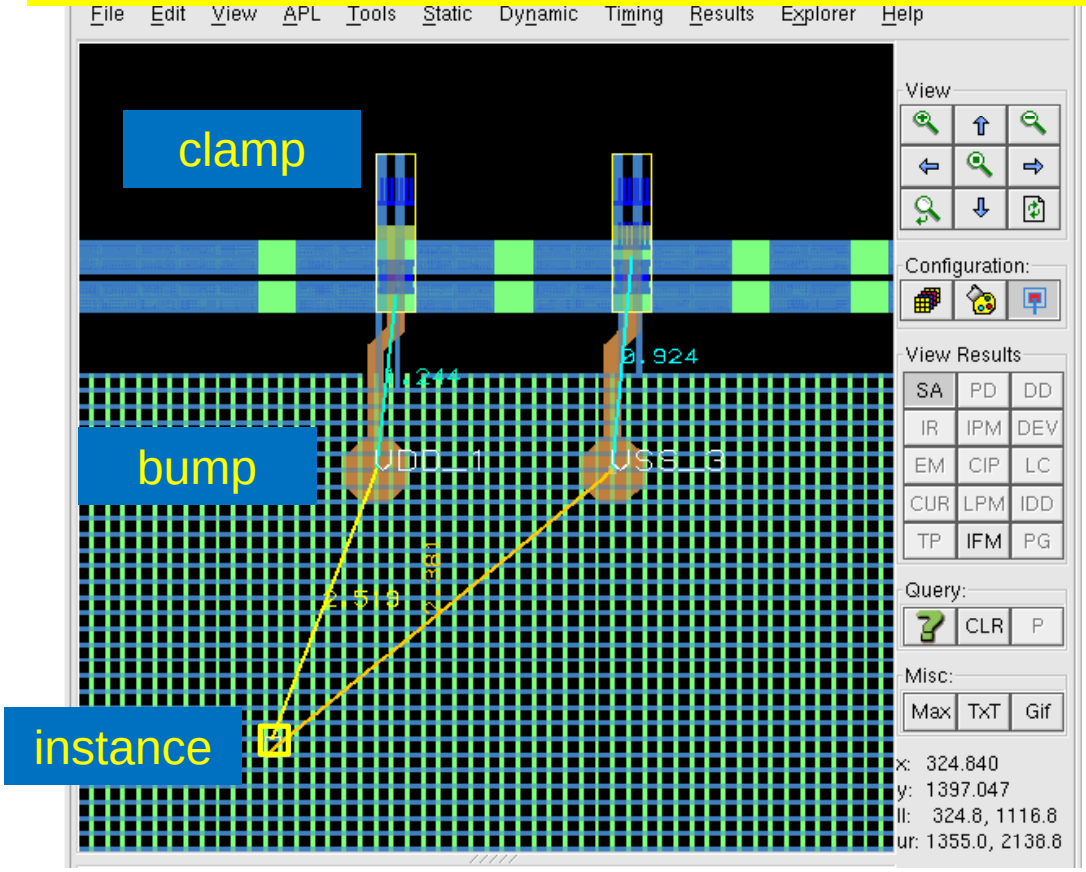
esd_show_worst (5/5)

I2B2C type (generated by 'esd_customize_report –f i2b2c')
inst-bump R and bump-clamp R are on each line



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esd_clear

- Purpose
  - ✓ To clear screen. (delete lines and makers, and clear selection)

- Usage
  - ✓ Use '-sa' option to reset all changes of zoom level, net selection, and layer selection

```
esd_clear
                ?-sa?

                -sa : show all nets and layers after clear screen

        <usage>
                esd_clear
                esd_clear -sa
```


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